

RESEARCH ARTICLE

Image Distinguishability Analysis Testing Through Principal Components and Its Application to Hot Spot Scale Invariance

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ABSTRACT

Hot spots are spatial regions of intense energy localization that govern initiation of secondary high explosives. Studies that characterize or compare simulated hot spots are frequently either qualitatively descriptive or resort to quantitative distribution functions that neglect stochastic variations and spatial correlations—effects that are also neglected in common comparison tests like the Kolmogorov–Smirnov test. To this end, we develop an image distinguishability analysis (IDA) test based on principal component (PC) analysis that makes pixel-by-pixel comparisons between small, for example, $O(<10)$, image data sets. The IDA test makes comparisons through a generalized distance metric in the PC space and a test statistic that is derived to calculate p -values. We derive a statistical distribution and criticality criterion to determine whether images are distinguishable from established baselines. We apply the IDA test on images generated from molecular dynamics simulations of hot spots from pore collapse in TATB to assess scale invariance in the complex patterns of hot spots that form in a representative high explosive crystal. The IDA test shows that TATB hot spot spatial temperature fields and their derived temperature histograms exhibit scale-invariant features over specific intervals of shock orientation, strength, and initial pore diameter. However, the IDA test also shows that qualitatively different conclusions regarding invariance can be reached depending on whether the hot spot is treated as a spatially correlated field as opposed to a distribution function that lacks spatial information.

1 | Introduction

Hot spots are regions of intense energy localization that accelerate chemical reactions responsible for secondary high explosive (HE) initiation [1]. Hot spots form when shock waves interact with material defects and they exhibit complicated structural features that depend on nonlinear coupling between geometric and material properties including anisotropic crystal mechanics, momentum and energy transport, and chemical reactions. Since the 1960s [2], physics-based computer simulations have played a leading role in understanding hot spot formation [3–5] and dynamical evolution [6–8]. Hot spot temperature fields have only been experimentally measured in the past few years [9,

10]. Most assessments of simulated and experimental hot spot images have relied on qualitative interpretations of temperature field patterns and simplified quantitative measures that neglect spatial correlations. Despite the growing body of data, gaps in understanding persist as most important metrics for HE safety and performance, such as the run distance to detonation or detonation failure diameters, have yet to be accurately predicted from the bottom up [11]. Recent advances in the direct comparison of simulated and measured temperature fields [12, 13] and in training machine learning models to computationally expensive physics simulations [11, 14] underscore the hot spot problem as a paragon for the rapidly growing need to rigorously quantify and compare small sets of images.

Images are the primary medium of many modern scientific analysis techniques, with wide-ranging contexts spanning computed tomography to characterize microstructure [15], measuring thermal [16] and mechanical [17] loads, and tracking reactive processes such as corrosion [18] to name only a few application areas within materials science beyond HE hot spots. The human eye is predisposed to detect real and imaginary patterns in images [19], but the rapid development of computer vision techniques has enormous potential to reduce human-in-the-loop efforts and biases in the interpretation of image data. While computer vision coupled with machine learning can rapidly compress, segment, classify, and generate synthetic images, these algorithms are often developed with large data sets in mind [20]. Moreover, the scope of these algorithms can be especially broad, such as classification algorithms that assign semantic identifiers to image contents (e.g., identifying all images depicting cats) [21]. These algorithms are powerful, but potentially disconnected from the goals of targeted scientific studies that involve (often dynamically evolving) structures and patterns where spatial correlations are important, but where only a small set of images are available due to high computational or experimental characterization costs. For instance, while the Eigenface approach, which is based on principal components (PCs), can compress and classify images of faces (or other objects), it is commonly applied to very large data sets, works in a feature space with hundreds of dimensions, and employs ad hoc thresholding for image classification [22, 23]. A comparison test that operates on small sets of images and is able to identify distinguishable differences beyond stochastic variations with corresponding p -values would help solve a wide range of problems.

To this end, we develop an image distinguishability analysis (IDA) testing framework that follows a similar compression methodology as the Eigenface approach [22, 23], but that works with smaller data sets and directly incorporates a test statistic for p -value testing. As discussed in detail below, the method makes pixel-by-pixel comparisons between samples of high-dimensional image-like data and computes a generalized distance metric in the PC space. Based on the hypothesis that inherent stochasticity is represented by independent Normal behavior of the PCs, we then derive a statistical distribution and corresponding critical value (CV) to decide whether images are stochastically similar or distinguishable from established baselines. As a concrete example for illustration, we apply the IDA test to temperature fields resulting from dynamical hot spot formation upon shock loading of an HE crystal.

Quantitative measures of images depicting field variables and statistical tests comparing these measures are cornerstones of physical science and the development of physical insights. For random variable X of a given field x with probability distribution $f(x)$, one of the most simple measures is the 1D empirical cumulative distribution function (CDF)

$$F_X(x) = \int_{-\infty}^x f_X(x') dx', \quad (1)$$

and the empirical complementary cumulative distribution function (CCDF)

$$\bar{F}_X(x) = 1 - F_X(x). \quad (2)$$

The CDF and CCDF, respectively, quantify the probability that X will take a value less than or greater than x . In the context of an HE hot spot, these measures serve as convenient means to simultaneously summarize the peak hot spot temperature and the amount of material above or below a given temperature, yielding the hot spot size.

Figure 1 shows three example 200×200 pixel images of hot spot temperature fields along with their respective empirical CDFs and CCDFs. As will be discussed at length below, all three hot spots resulted from shocking microstructural pore defects with equivalent diameter D and particle velocity u_p , but C resulted from a different shock direction relative to the HE crystal compared to A and B. It is immediately apparent from the images that hot spots A and B are quite similar to each other and that both are qualitatively different from hot spot C. A logarithmic plot of the CCDF is particularly useful in the context of hot spots as it allows one to quickly determine the peak temperature, which is, respectively, 984, 1130, and 1024 K for hot spots A, B, and C. While the CDF shows that A and B temperature distributions are quite similar, it is perhaps surprising that the log plot of the CCDF reveals A and C to have more similar high-temperature tails as compared to hot spot B. The high-temperature tail is of particular relevance to hot spot evolution, as HE chemical reactions generally follow Arrhenius temperature laws that are sensitive to small changes in the peak temperature [6]. For this reason, temperature distribution functions and their CCDFs have found application in a large body of hot spot studies by different groups in the literature [4, 5, 12, 14, 24–29], including our own work [30].

Direct comparison tests on images are increasingly relevant as the physical accuracy and fidelity of numerical simulations improves and as experiments become increasingly capable of spatially resolving features. Such tests form the basis for assessing the accuracy of coarse-grained simulations [27], the training and validation of machine learning models [14], and in the direct comparison of simulations and experiments [12, 13]. The root mean-square difference (RMSD) is perhaps the most rudimentary pixel-by-pixel comparison one could make between images. However, the resulting value offers limited physical insight, particularly with respect to distribution tails. Full 2D maps of image differences provide more detailed spatial views [8], but what constitutes a *large* difference remains open to subjective interpretation. A more substantial comparison would seek to define an unambiguous statistical criterion for whether images can be distinguished.

The Kolmogorov-Smirnov (KS) test is a popular textbook hypothesis-testing framework to determine whether two samples (e.g., images) are drawn from the same distribution [31]. Briefly, the KS test defines a distance metric in terms of the maximum difference between the sample CDFs

$$d_{KS} = \max_x |F_1(x) - F_2(x)|, \quad (3)$$

where $F_1(x)$ and $F_2(x)$ are the empirical CDFs for samples 1 and 2. The p -value that the null hypothesis is true—that is, that the two samples are drawn from the same distribution—for a given d_{KS} can then be determined from the KS distribution.

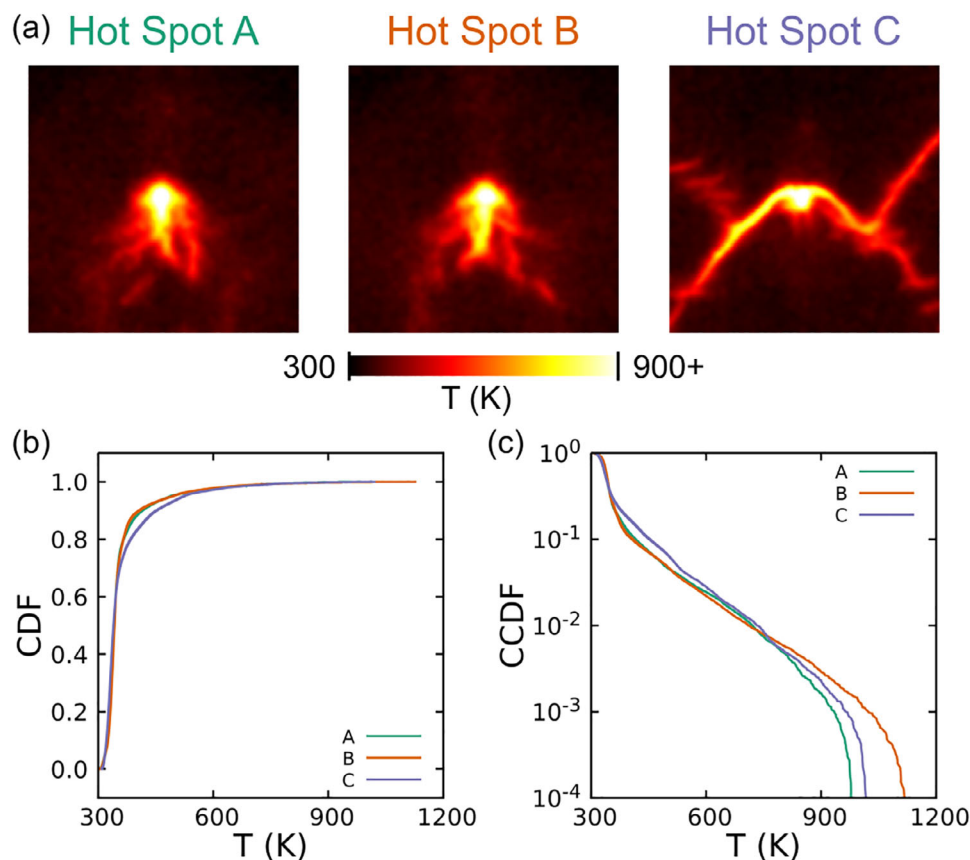


FIGURE 1 | (a) Example images of hot spot temperature fields in shocked HE crystal resulting from the collapse of a cylindrical pore with diameter D . Image fields of view span an area that is $2D \times 2D$. Simple statistical measures of each hot spot temperature field including (b) the empirical CDF and (c) the empirical CCDF. Note the logarithmic scale on the CCDF.

TABLE 1 | Simple statistical comparisons of the hot spot fields in Figure 1.

Pair	RMSD (K)	d_{KS}	KS p -value
A–B	21.5	3.3×10^{-2}	1.1×10^{-18}
A–C	72.3	1.0×10^{-1}	9.4×10^{-186}
B–C	70.7	1.3×10^{-1}	2.5×10^{-280}

Smaller p -values indicate greater confidence in rejecting the null hypothesis. A decided advantage of the KS test is that it is translationally and rotationally invariant, reducing high-dimensional field data to a 1D comparison of distributions. While other distance metrics can be more robust to spatial information loss such as the Wasserstein distance, they often lack a known distribution function for calculating p -values (unlike the KS distance).

Table 1 shows results from application of the above standard comparison metrics to the three pairs of hot spot images. Similarity between A and B is evident from the comparatively small RMSD, which is more than a factor of three smaller than the RMSDs for pairs A–C and B–C. However, the KS distances for all three pairs have associated p -values that are vanishingly small. It is evident that pairs A–C and B–C are more dissimilar than pair A–B, but the KS test clearly shows that hot spots A and B can be distinguished

from each other. This result that A and B are distinguishable is problematic in that both hot spots resulted from macroscopically equivalent conditions; the only functional difference is that they were obtained from independent all-atom simulations starting from different sets of atomic velocities. A physical interpretation within the context of statistical mechanics suggests that the variance between A and B reflects the typical level of microscopic variation for quantities that ought to be considered the same from a macroscopic perspective. For this reason, a pixel-by-pixel comparison test that accounts for physically meaningful variations between samples drawn from the same distribution is desirable.

In what follows, we develop a general IDA testing approach that is capable of making pixel-by-pixel comparisons between samples of image-like data. Our proposed test builds on earlier work [32] that demonstrated the use of PC score clustering according to microstructural similarities based on summary features computed for sample images. The present work involves several key extensions, including (1) a generalized distance metric and corresponding hypothesis test, and (2) that it works to directly compare images rather than summary features. These extensions enable the IDA test to identify when images are distinguishable from established baselines while accounting for typical fluctuations inherent to a given (and generally unknown) distribution governing those baselines. As an example, we apply the IDA test to reexamine our recently reported [30] micron-scale all-atom

TABLE 2 | Definitions for the six baseline shock conditions.

Shock condition	D (nm)	u_p (km/s)	θ (deg.)
A_1	60	0.5	0
A_2	60	0.5	90
A_3	60	1.0	0
A_4	60	1.0	90
A_5	60	2.0	0
A_6	60	2.0	90

molecular dynamics (MD) simulations to build a firmer case for scale invariance of the hot spot formation process. General requirements for applying the IDA test and a prospectus for its use in other materials science problem areas are summarized at the end.

2 | MD Simulations

All images analyzed here are temperature field results from large-scale classical MD simulations that were performed using the LAMMPS code [33]. Each MD simulation used a 3D periodic simulation cell and a validated variant [34] of the nonreactive force field (FF) for the HE TATB (1,3,5-triamino-2,4,6-trinitrobenzene) originally developed by Bedrov et al. [35]. Shock-induced hot spot formation was modeled using a reverse-ballistic setup [36] and quasi-2D simulation cells of an oriented TATB crystal sample containing a cylindrical pore of diameter D . The simulation geometry models a supported shock generated by an infinitely massive flyer that impacts the sample at prescribed velocity u_p (i.e., the particle velocity). Resulting collapse of the pore leads to localization of energy in a hot spot. Using a mechanically accurate and computationally efficient nonreactive FF enables performing many independent MD simulations to sample variance in the initial hot spot temperature fields that form during pore collapse, but these simulations do not accurately capture long-time hot spot evolution because they neglect chemical reactions. More detailed discussion on MD simulation algorithms [37–41] and trajectory analysis can be found in Ref. [30].

Six unique shock conditions A_k were considered in which we varied the shock direction $\mathbf{S}(\theta)$ relative to the crystal and the shock strength through u_p for fixed pore diameter $D = 60$ nm (see Table 2). Two directions were considered that, respectively, oriented the shock along ($\theta = 0^\circ$) or normal ($\theta = 90^\circ$) to the TATB crystal layers and three shock strengths were chosen that span experimental thresholds [42] for TATB shock initiation. Size scaling effects were assessed by scaling the pore diameter D , which was varied between 10 and 300 nm, corresponding to systems with between 0.7 and 600 million atoms. Hot spot temperature fields were obtained as scaled square 200×200 pixel two-dimensional images through postprocessing of the all-atom trajectories. The field of view spanned an area with dimensions $2D \times 2D$ and was centered on the core of the hot spot.

3 | Reduced Dimensional Representation

It is straightforward to define and compute a generalized distance between two images in terms of pixel-by-pixel differences. However, as mentioned in the forgoing discussion, interpreting the distance in terms of physically meaningful distinguishability is complicated due to the presence of (1) natural stochastic variations in intensity, and (2) strong spatial correlations of pixel intensities. To address the first issue, we performed four independent MD simulations for each of the six shock conditions A_k listed in Table 2, with each simulation varying only the initial thermal atom velocities, which are stochastic in nature. This yielded a set of 24 “baseline” images, with four images for each shock condition. To address the second issue, we made the (testable) hypotheses that when images are represented in the reduced-dimensional PC space, the resulting PC scores satisfy three properties, that is, (a) they cluster according to shock condition, (b) within a given shock condition the image scores for each PC component follow the Normal distribution, and (c) the scores for different PC components are statistically independent (i.e., uncorrelated). Toward the end of the present section, we show that property (a) is satisfied, that is, the PC scores indeed cluster. In the next section, we address (b) and (c), where we show that those two properties make it possible to define a generalized distance metric that follows a known statistical distribution, which enables hypothesis testing and p -value computation to determine if new images are distinguishable from baseline images belonging to a given condition. In what follows, we develop the necessary mathematical framework for the IDA test using the hot spot problem as an example. A Python code implementation is provided in the [Supporting Information](#).

Figure 2 displays all $n = 24$ baseline hot spot images, with four each for the six unique shock conditions A_k in Table 2. As evident from the figure, images with the same shock condition are similar to each other but are dissimilar from images with different shock conditions. Specific structural features of the hot spots will be discussed in a later section where we apply the IDA test to hot spot scale invariance.

Let each 2D image i be represented as a $p \times 1$ column vector $\mathbf{I}_i$, where the p components are the intensities (e.g., temperature values) at each of the p pixels. Note that this definition of an image is highly general. An image might be comprised of a single rasterized 2D field variable such as the temperature, include multiple appended field variables, or even contain a time series of one or more field variables. Furthermore, there is no requirement that the image data be 2D in nature, and instead could be of lower or higher dimensionality (e.g., 0D scalar, 1D spectrum, or ≥ 3 D voxel grid). The salient requirement is that each image has a consistent rasterization scheme such that vectorization of the data leads to a consistent pixel-by-pixel mapping. For the purpose of exposition, we consider an analysis with $n = 24$ baseline images of the $p = 200 \times 200$ pixel temperature field.

The first step in the IDA test is to establish a basis set of images to represent high-dimensional image data in terms of a much smaller number of basis projection scores (i.e., PC scores). We accomplish this using the same general PC-based compression procedure as the Eigenface method [22, 23], but where we work

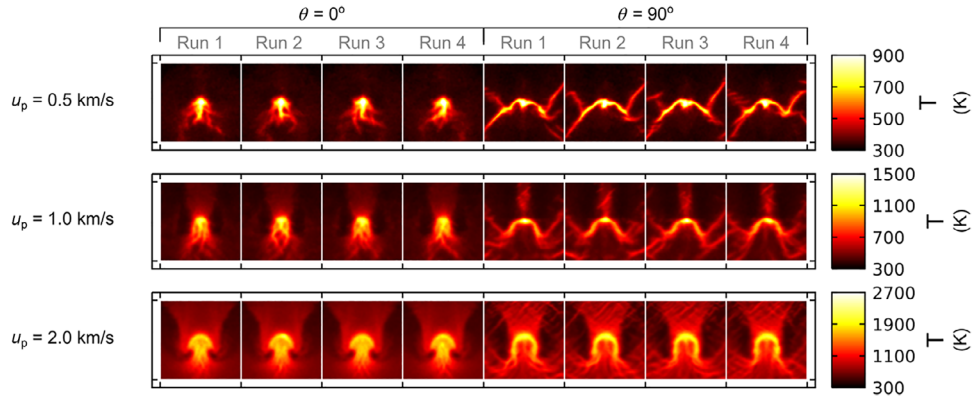


FIGURE 2 | Images of the 24 baseline hot spot temperature fields obtained with 60 nm pores. Four independent runs were performed for each of the six unique (θ, u_p) conditions $A_1, \dots, A_6$.

in a (generally much smaller) feature space spanned by the set of baseline images. To this end, the mean image

$$\bar{\mathbf{I}} = \frac{1}{n} \sum_{i=1}^n \mathbf{I}_i \quad (4)$$

is computed and subtracted from each image to obtain a set of mean-centered images $\mathbf{I}_{ic} = \mathbf{I}_i - \bar{\mathbf{I}}$. These mean-centered images are collected into an $n \times p$ matrix

$$\mathbf{M}_c = [\mathbf{I}_{1c} | \mathbf{I}_{2c} | \dots | \mathbf{I}_{nc}]^T, \quad (5)$$

in which the i th row contains the pixels of mean-centered image $\mathbf{I}_{ic}$. Note that the rank of $\mathbf{M}_c$ is $r = n - 1$ because the sum of each column of $\mathbf{M}_c$ is zero due to the mean centering.

Diagonalizing the $p \times p$ matrix $\mathbf{M}_c^T \mathbf{M}_c$ yields the PCs as the $r = 23$ eigenvectors corresponding to the r nonzero (and positive) eigenvalues $\{\lambda_1, \lambda_2, \dots, \lambda_r\}$ ordered from largest to smallest in value. The relative weight (or importance) of PC j is computed as

$$f_j = \frac{\lambda_j}{\sum_{k=1}^r \lambda_k}. \quad (6)$$

The corresponding eigenvectors $\mathbf{e}_j$ are the so-called loading vectors, where $\mathbf{e}_j$ is the $p \times 1$ loading vector of the j th PC. Figure 3 shows image representations of the 23 PC eigenvectors obtained from the 24 baseline images in Figure 2 along with their relative importance. In the present case, 89.4% of the image data is captured by PC 1, 6.1% is captured by PC 2, and 2.2% is captured by PC 3, with the remaining PCs having even smaller weights. The first two PCs account for over 95% of the data, and the first seven account for over 99%. It is also clear that the PCs with the greatest weights exhibit features that directly connect to the structural characteristics in the hot spot images whereas the lower-weight PCs are largely noise.

The projection of mean-centered image $\mathbf{I}_{ic}$ on eigenvector $\mathbf{e}_j$ is the “score” of image i along the j th PC. Thus, from the $p \times r$ loadings matrix

$$\mathbf{L} = [\mathbf{e}_1 | \mathbf{e}_2 | \dots | \mathbf{e}_r], \quad (7)$$

one can compute the $1 \times r$ score vector $\mathbf{x}_i$ of a given image $\mathbf{I}_i$ as

$$\mathbf{x}_i = \mathbf{I}_{ic}^T \cdot \mathbf{L}, \quad (8)$$

such that x_{ij} is the score of mean-centered image i along the j th PC. Note that scores can be used to reconstruct any of the n images as

$$\mathbf{I}_i = \bar{\mathbf{I}} + \sum_{j=1}^r x_{ij} \mathbf{e}_j. \quad (9)$$

An example of this reconstruction property is given in Figure 4, which shows the iterative corrections to the mean image $\bar{\mathbf{I}}$ as one sums over the factors $x_{ij} \mathbf{e}_j$ for each of the 23 PCs. It should be noted that the PCs form just a complete enough set to cover the space of the n mean-centered images. A general image can be compressed and reconstructed through such projections, but with the understanding that there will be some loss due to incompleteness of the PC basis set.

The primary advantage of the above PC analysis is that it reduces the complexity of comparing full p -pixel images to an assessment in a lower-dimensional space while still representing a pixel-by-pixel comparison. Images belonging to the six different shock conditions can be distinctly classified based on comparisons of their PC scores. Figure 5 shows plots of the PC scores for the 24 images from Figure 2 with symbols denoting the corresponding shock condition for which a given image is a baseline. As can be clearly seen, images belonging to the six conditions cluster tightly together in the various PC subspaces. This tracks with the notion that images of a given shock condition are “similar” to each other. More importantly, this clustering serves as a foundation for making statistical comparisons of new images to those representing a baseline for a particular shock condition.

4 | PC-Score-Based Distance Metric

Having established property (a), that is, PC scores of similar images cluster together (see Figure 5), we now turn our attention to properties (b) and (c) outlined in the previous section. These properties concern the nature of the variation of the PC scores of images within each cluster. Both these hypotheses are central to

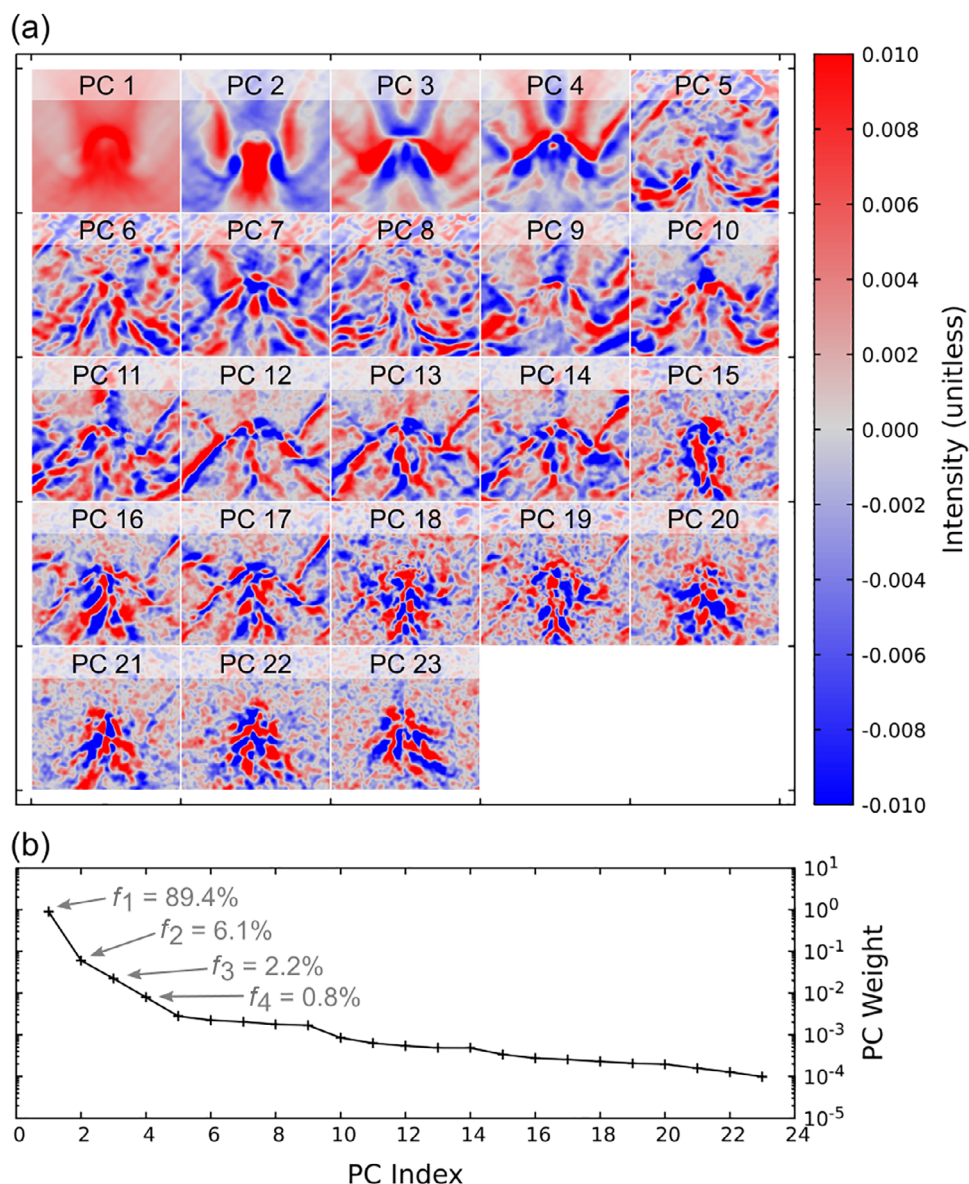


FIGURE 3 | (a) Image representations of the 23 PCs obtained from the baseline images in Figure 2. (b) Weight of each PC as computed from Equation (6).

the construction of an appropriate distance metric for the IDA test and are stated as:

- **Hypothesis 1**, for an image i belonging to shock condition A , its PC score x_{ij} (where j is PC index) follows the Normal distribution $N(\mu_{Aj}, \sigma_{Aj}^2)$; and
- **Hypothesis 2**, for any image i , the scores of different PC components are statistically independent, that is, $x_{ij} \perp x_{ik}$ for $j \neq k$.

At the end of this section, we show strong statistical evidence in support of both the above hypotheses for the system of 24 images considered here. For now, we simply assume their validity and proceed to formulate a distance metric that follows a known distribution.

Given Hypothesis 1, a set of n_d images within a given condition A can be considered as a sample of scores from a Normal population (for each PC component j) with sample mean and sample variance defined, respectively, as

$$\bar{x}_{Aj} = \frac{1}{n_d} \sum_{i \in A} x_{ij}, \quad (10)$$

and

$$s_{Aj}^2 = \frac{1}{n_d - 1} \sum_{i \in A} (x_{ij} - \bar{x}_{Aj})^2, \quad (11)$$

where the sum runs over the $n_d = 4$ baseline images belonging to shock condition A .

Consider a new image δ with PC score vector $x_{\delta j}$ for $j = \{1, \dots, r\}$. If the image belongs to shock condition A , then it is

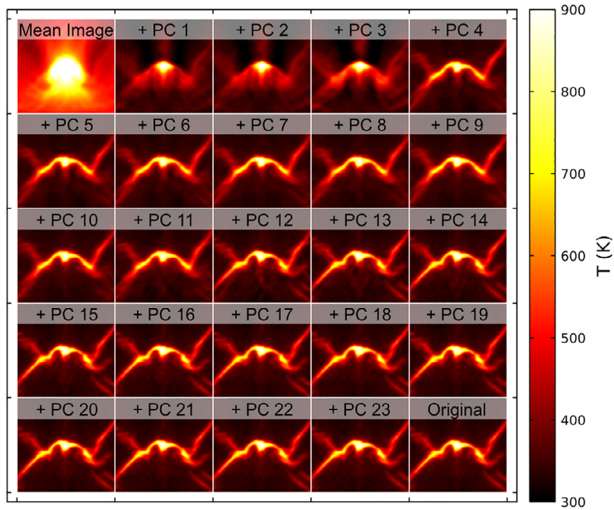


FIGURE 4 | Example reconstruction of the hot spot image from Run 1 with $\theta = 90^\circ$ and $u_p = 0.5$ km/s given in Figure 2 using the PC scores. Each panel shows the result from the running sum in Equation (9) as contributions from each PC are successively added to the mean image.

straightforward to show that under Hypothesis 1 its normalized score $z_{\delta j} = (x_{\delta j} - \bar{x}_{A_j})/s_{A_j}$ follows the distribution $z_{\delta j} \sim \sqrt{(n_d + 1)/n_d} t_{n_d-1}$ where t_{n_d-1} denotes the Student's t distribution with $df = n_d - 1$ degrees of freedom. We note that this result is valid when image δ is *not* used to compute the mean and variance using Equations (10) and (11).

Given the above result, we propose the following L_2 squared-distance metric

$$d_{\text{IDA}}(\mathbf{I}_\delta, A) = \sum_{j=1}^r f_j \frac{(x_{\delta j} - \bar{x}_{A_j})^2}{s_{A_j}^2} \quad (12)$$

for image δ from baseline images belonging to condition A . Based on the distribution of $(x_{\delta j} - \bar{x}_{A_j})/s_{A_j}$ as discussed above, if we also assume Hypothesis 2 to be valid, then d_{IDA} should follow the distribution

$$d_{\text{IDA}}(\mathbf{I}_\delta, A) \sim \frac{n_d + 1}{n_d} \sum_{j=1}^r f_j t_{j, n_d-1}^2, \quad (13)$$

where each t_j on the right-hand side (RHS) is an independent t variable with the same $df = n_d - 1$. The design of Equation (12) ensures two important properties of the IDA metric: (1) it accounts for spatial correlation of pixel intensity within images; and (2) it boosts denoising effects. Property (1) follows from the fact that the PCs themselves incorporate statistically independent linear combinations of the pixel intensities, and the images in turn can be expressed as a linear combination (i.e., scores) of the PCs. The independence of the PC scores is what enables the statistical distribution of d_{IDA} to be expressed as a simple sum over the PCs (Equation 13). If the distance metric were formulated within the original spatial pixel space instead of the PC space, the resulting statistical distribution would involve a large $p \times p$ covariance matrix, which becomes a much more challenging problem, especially with so few images available for training. Property (2) results from the weight factors f_j , which

are designed to enhance the importance of the first few PCs that encode the most important condition-to-condition variations while suppressing high-index PCs, which mostly describe noise. This reduction in noise also appears to lead to a consistent choice in p -value as a quantitative criterion for image distinguishability, as demonstrated in the analysis to follow. More specifically, uniform weighting of the contributions of each PC (i.e., setting $f_j = 1$) can lead to inconsistencies in the interpretation of results, as shown later in this section and in the SI.

To decide whether image δ is distinguishable (or not) from condition A one needs to compare $d_{\text{IDA}}(\mathbf{I}_\delta, A)$ with the CV of the weighted-sum- t^2 distribution on the RHS of Equation (13), or equivalently, compute the corresponding p -value for the $d_{\text{IDA}}(\mathbf{I}_\delta, A)$ statistic. Within standard statistical conventions, small p -value (e.g., <0.001) should imply a high degree of distinguishability of test image δ from condition A , while a large p -value (e.g., >0.1) should be interpreted as image δ being indistinguishable from condition A . Note that a nonweighted ($f_j = 1$) sum of independent t^2 variates follows an F distribution, but a weighted sum of t^2 (as in Equation 13) does not follow any closed-form, standard statistical distribution. Thus, to compute the CV and p -value, we performed Monte Carlo (MC) simulations in which we repeatedly draw a random sample of r independent t -variates (with $df = n_d - 1$) a large number ($n_{\text{MC}} = 10^5$) of times and compute the resulting statistic by using the formula on the RHS of Equation (13). The corresponding CV at a level of significance α is then obtained as the $100(1 - \alpha)$ th percentile from the resulting MC set of n_{MC} values of the statistic, and the p -value is obtained as $1 - q$, where q is the quantile position of $d_{\text{IDA}}(\mathbf{I}_\delta, A)$ in the above MC set. Several repeats of the above procedure confirmed numerical stability of the procedure, with a standard error for CV or p -value being less than just a few percent.

The top panel of Figure 6 displays six new independent “test” images, one each for the six different shock conditions $A_1, \dots, A_6$, while the bottom panel plots the corresponding d_{IDA} distances (in log scale) of these new images from the baseline clusters of each of the six shock conditions. The solid horizontal line, which indicates the MC-computed CV at $\alpha = 0.01$ cleanly separates the intracondition d_{IDA} distances from intercondition distances, thus showing consistency of the proposed d_{IDA} metric and the associated statistical distribution. The two dashed lines indicate the CV values corresponding to $\alpha = 0.001$ and $\alpha = 0.1$, respectively. We see that at $\alpha = 0.001$ (upper dashed line) the CV is close to several intercondition d_{IDA} distances, although it is still able to correctly classify distinguishability of all test images from interconditions. In other words, all intercondition p -values were 0.001 or less. However, at $\alpha = 0.1$ (lower dashed line) some of the intracondition indistinguishability is falsely classified, with the lowest p -value being ≈ 0.04 (d_{IDA} for test image 2 from condition 2). Such exploration sets some practical guidelines on the levels of significance α to be employed in the use of the IDA method in different applications. For our problem at hand, a good choice appears to be $\alpha \in (0.001, 0.04)$. While the above analysis computed d_{IDA} using all $r = 23$ PCs, essentially identical classification results are obtained in the present example when using only the two most important PCs.

At this juncture, we point out the importance of the weight factors in the formula for d_{IDA} (Equation 12). If we use uniform

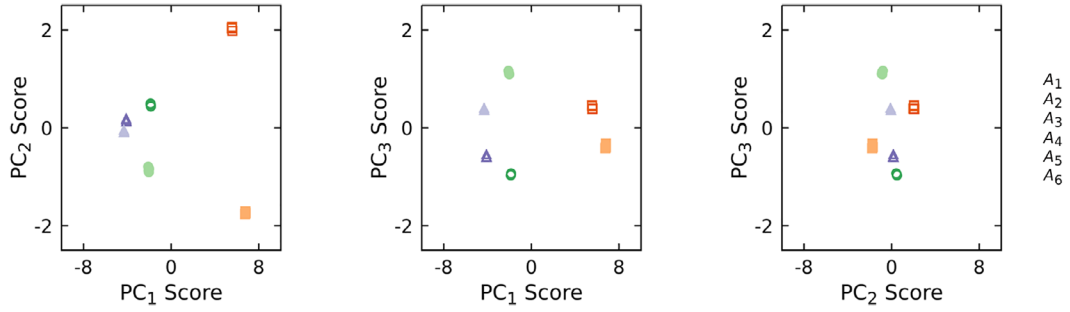


FIGURE 5 | PC scores for the 24 images in Figure 2 plotted in different PC subspace planes. Independent images belonging to the same shock condition are plotted using the same symbol. Note that all PC scores have been scaled by 1/10 000 to simplify the plotting axes.

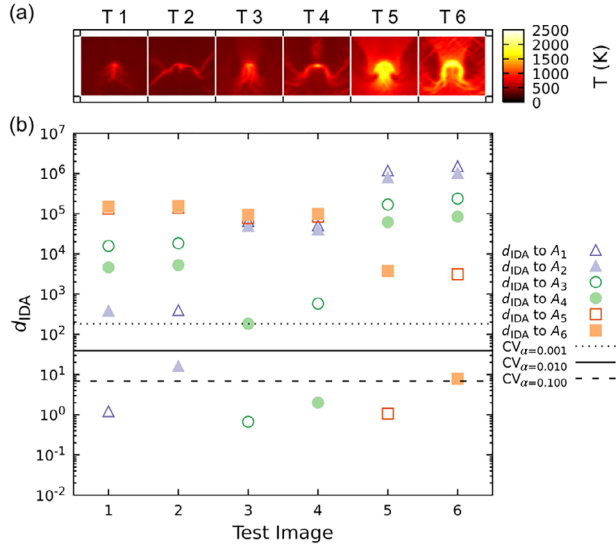


FIGURE 6 | (a) Images of six new test images $\mathbf{I}_\delta$ obtained with 60 nm pores, with one performed per unique shock condition A_k . (b) Classification of six new test images based on the $d_{\text{IDA}}(\mathbf{I}_\delta, A_k)$ distance metric in Equation (12) and the computed d_{IDA} CV for selected α .

weight for all PCs, we do not find a single value of α where the corresponding CV separates the intercondition d_{IDA} from the intracondition d_{IDA} simultaneously for all six test images (see Figure 1, SI). From the d_{IDA} values in Figure 6, we also see the relative closeness of condition pairs (A_1, A_2) , (A_3, A_4) , and (A_5, A_6) , that is, patterns within the same piston speed (u_p) but different shock directions (θ) (see Table 2) are more similar than patterns resulting from the different piston speeds considered in this work. Such insights can be very useful in interpreting results from targeted experiments or grain-scale shock simulations on realistic microstructures consisting of multiple defect sites for potential hot spots.

We end this section with a more concrete justification of the two hypotheses that led to Equations (12) and (13).

Hypothesis 1: Verification that $x_{ij} \sim N(\mu_{A_j}, \sigma_{A_j}^2)$. Since we do not know the population parameters μ_{A_j} and $\sigma_{A_j}^2$, we compute the sample estimates $\bar{x}_{A_j}$ and $s_{A_j}^2$ using the $n_d = 4$ sample images within each class. We then normalize each score as $z_{ij} = (x_{ij} - \bar{x}_{A_j})/s_{A_j}$. If we consider the first n_{PC} PCs, the 24 baseline images lead to a collection of $24n_{\text{PC}}$ normalized z_{ij} scores. To show that

they are consistent with Hypothesis 1, we create a set of four scores by independent draws from a standard Normal distribution $N(0, 1)$, compute their mean $\bar{x}$ and sample standard deviation s , and normalize them as $(x - \bar{x})/s$. Repeating this process $6n_{\text{PC}}$ times results in a reference set of $24n_{\text{PC}}$ numbers. Figure 7a shows a quantile–quantile (QQ) plot comparing z_{ij} scores with the reference set for $n_{\text{PC}} = 6$. A high degree of parity is clear, which is strong evidence that the PC scores x_{ij} indeed follow a Normal distribution.

Hypothesis 2: Verification that $x_{ij} \perp x_{ik}$ for all images i and all PCs $j \neq k$. Given that x_{ij} and x_{ik} each appear to follow a Normal distribution as established above, their mutual independence implies that the sample correlations $r_{jk} = \text{cov}(x_{ij}, x_{ik})/(s_{x_{ij}} s_{x_{ik}})$ will follow

$$r_{jk} \sim \frac{t_{n_d-2}}{\sqrt{t_{n_d-2}^2 + n_d - 2}} \quad (14)$$

for all $j \neq k$ where t_{n_d-2} is the Student t distribution with $\text{df} = n_d - 2$ degrees of freedom. To check that Equation (14) holds, we compute the sample correlation coefficient r_{jk} for all $j < k \in \{1, 2, \dots, n_{\text{PC}}\}$ within each class, which leads to a collection of $N = 6n_{\text{PC}}(n_{\text{PC}} - 1)/2$ distinct correlation values. An analogous set of N random variates r_t are drawn from a t distribution with $\text{df} = n_d - 2$, which are rescaled by $y_t = r_t / \sqrt{r_t^2 + n_d - 2}$. Figure 7b shows QQ plots comparing these two sets $\{r_{jk}\}$ and $\{y_t\}$ using different numbers of PCs n_{PC} . Parity is clear from a comparison against the $y = x$ line, which is strong evidence that the PC scores x_{ij} and x_{ik} are independent for $j \neq k$. It should be noted that the QQ results can fluctuate due to run-to-run variation for different random t_{n_d-2} draws. Such fluctuations can be large, particularly for smaller number of PCs such as $n_{\text{PC}} = 4$ and 6, which, respectively, show $N = 36$ and 90 points.

5 | Hot Spot Scale Invariance

5.1 | General Features

Having established the IDA procedure on a statistical footing, we now apply it to hot spot images of different pore dimensions to see if it can objectively quantify the presence of interesting patterns such as scale invariance in the temperature distribution fields around the core. Recent MD simulations of TATB hot spot formation yielded tentative evidence for temperature field

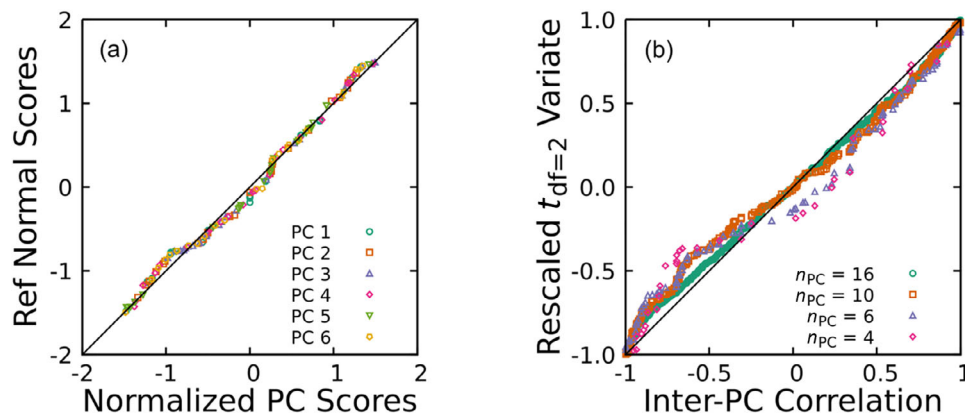


FIGURE 7 | (a) QQ plot of normalized PC scores relative to normalized scores of $N(0, 1)$ variates for $n_{PC} = 6$. (b) QQ plot of inter-PC score correlation relative to rescaled random t variates ($df = n_d - 2$) shown for different number of PCs considered, including $n_{PC} = 4, 6, 10$, and 16 .

scale invariance for pore diameters $D > 40$ nm when the shock pressure is $P \leq 8$ GPa [30] which is below the experimental initiation threshold for typical TATB-based HE formulations [42]. These determinations were based on an inspection of the temperature field CCDFs, which as demonstrated in the introduction has shortcomings when applied to comparisons of hot spot fields. Here, we reexamine these MD simulations using the IDA test and perform new MD simulations probing stronger shocks above the initiation threshold to establish firmer evidence for scale invariance.

Figure 8 shows resulting scaled images of hot spot temperature fields that were calculated as a function of pore diameter $D \leq 100$ nm for the six different orientation and piston velocity cases (θ, u_p) listed in Table 2. In some of our analyses below, we also consider four additional simulations with $D = 200$ and 300 nm that were performed for the 90° case with $u_p = 0.5$ and 1.0 km/s; images of these simulations can be found in Ref. [30]. Approximate shock pressures for the three u_p cases $0.5, 1.0$, and 2.0 km/s are, respectively, $2, 8$, and 22 GPa. Results for a given u_p were taken at equivalent times $\tau^* = \tau/D$, where τ corresponds to the time required for the shock to transit the entire compressible sample and reach the free surface. We chose $\tau^* = 1.50, 1.25$, and 0.92 ps/nm for $u_p = 0.5, 1.0$, and 2.0 km/s, respectively. These times are sufficiently long to attenuate mass flow from pore collapse and sufficiently short to avoid interactions of the resulting pressure pulse through the periodic boundary and cooling of the hot spot through thermal diffusion [8].

Inspection of the images shows a high degree of similarity between simulations for a given (θ, u_p) with different D , especially for $D \geq 40$ nm. As was previously noted for the two weaker shock cases [30], the larger simulations exhibit fine structure as the system size is scaled up, but the qualitative features are consistent across the different sizes. A similar size scaling result holds true for the new $u_p = 2.0$ km/s cases.

The 2D hot spot structure is a strong function of both θ and u_p . Hot spots obtained with the 0° orientation are roughly ellipsoidal with a hot central core region whereas the 90° orientation cases exhibit more complicated shear banding structures. As a purely qualitative and heuristic assessment, differences between the orientation cases appear smallest for the strongest shock strength.

At $u_p = 2.0$ km/s, both the 0° and 90° cases exhibit an arched shape with the main difference being that the 0° cases exhibit a hot central core whereas the 90° cases exhibit a thin horse-shoe-like region of high temperature. Many of these qualitative differences have been noted in previous MD simulations of TATB for smaller system sizes [5, 29].

Figure 9 shows empirical CCDFs for the hot spot images, which quantitatively corroborate the apparent similarities seen in the images. For example, the fact that the smallest pore diameters lead to colder hot spots is readily apparent. In most (θ, u_p) cases, both the 10 and 20 nm pores produce peak hot spot temperatures that are hundreds of Kelvin colder than obtained with $40+$ nm pores. Moreover, the 40 and 100 nm CCDFs (for values $> 10^{-2}$) are generally within the spread of the five independent 60 nm simulation results. Only the $(0^\circ, 2.0$ km/s) case differs from these trends in that the CCDFs for the 10 and 20 nm pores are quite similar to their larger counterparts. While these CCDFs provide evidence that TATB hot spots are invariant to system size for pores ≥ 40 nm, this comparison does not prove that the resulting temperature fields are scale invariant. Determining whether full images of the hot spot temperature field obtained for $D \neq 60$ nm are distinguishable from the $D = 60$ nm baselines for a given θ and u_p would provide much stronger evidence in support of scale invariance in hot spot formation.

5.2 | IDA Test Results

In formulating the IDA test, we showed above in connection with Figure 6 that it correctly classifies test images of hot spots to baseline shock conditions when applied to pores of equivalent diameter to the baseline condition ($D = 60$ nm). Here, we apply the IDA test to the hot spot images in Figure 8 obtained for pores with different diameters and compare them against the $D = 60$ nm baselines. This comparison measures whether changes in hot spot image patterns induced by changing the pore diameter are indistinguishable from the inherent stochastic variations established for a fixed diameter case. Thus framed, the IDA test can establish pore diameter intervals over which the hot spots are indistinguishably similar, providing more rigorous statistical evidence for scale invariance in hot spot formation.

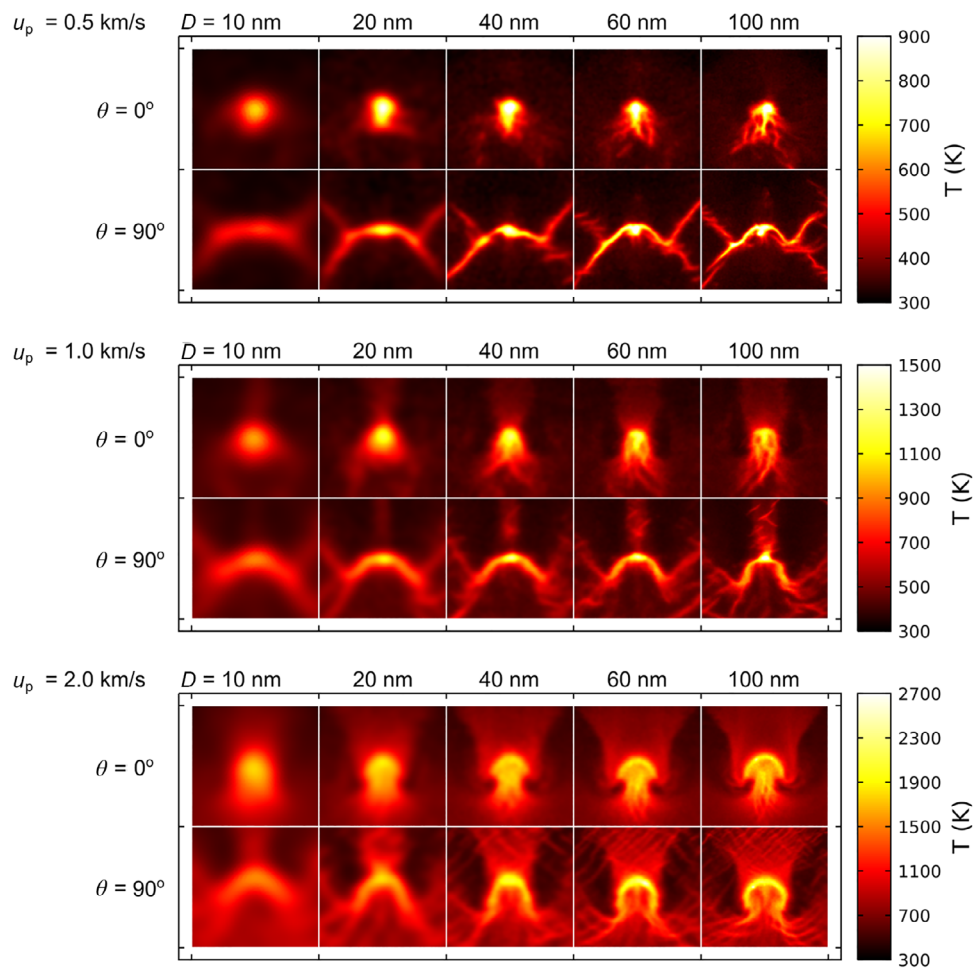


FIGURE 8 | Comparison of hot spot images for different pore diameters D at selected fixed θ and u_p . Each image is 200×200 pixels, shows an area that is $2D \times 2D$, and is centered on the hot spot core.

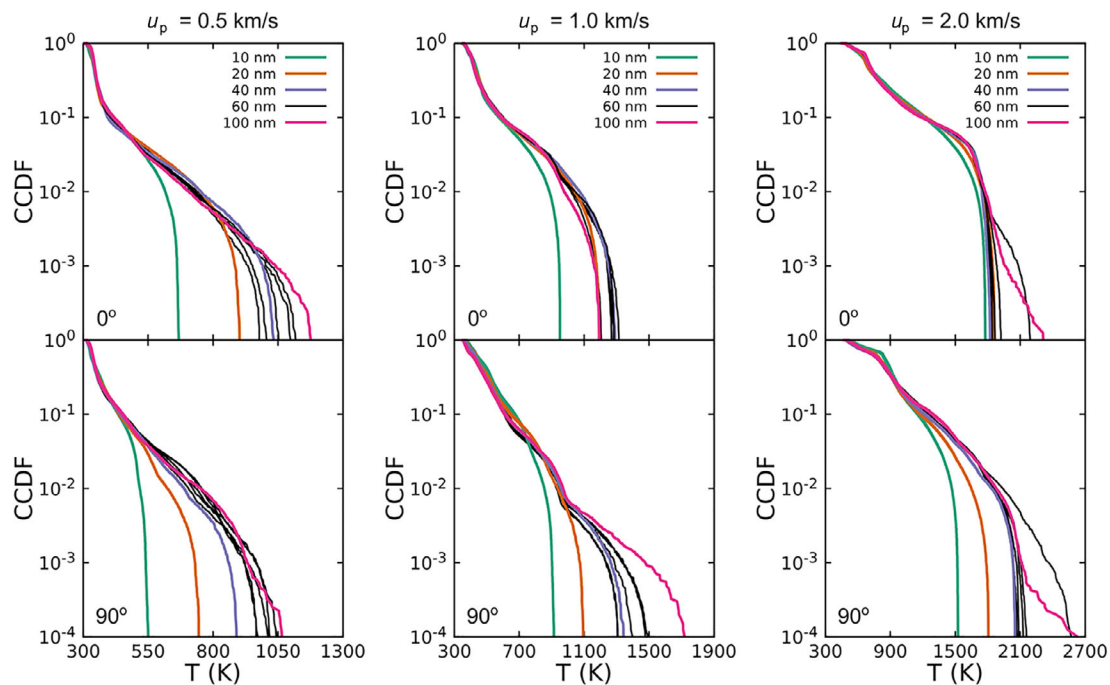


FIGURE 9 | Comparison of hot spot temperature empirical CCDFs computed for different pore diameters D at selected fixed θ and u_p . All five independent simulations for each 60 nm case are plotted as thin black curves.

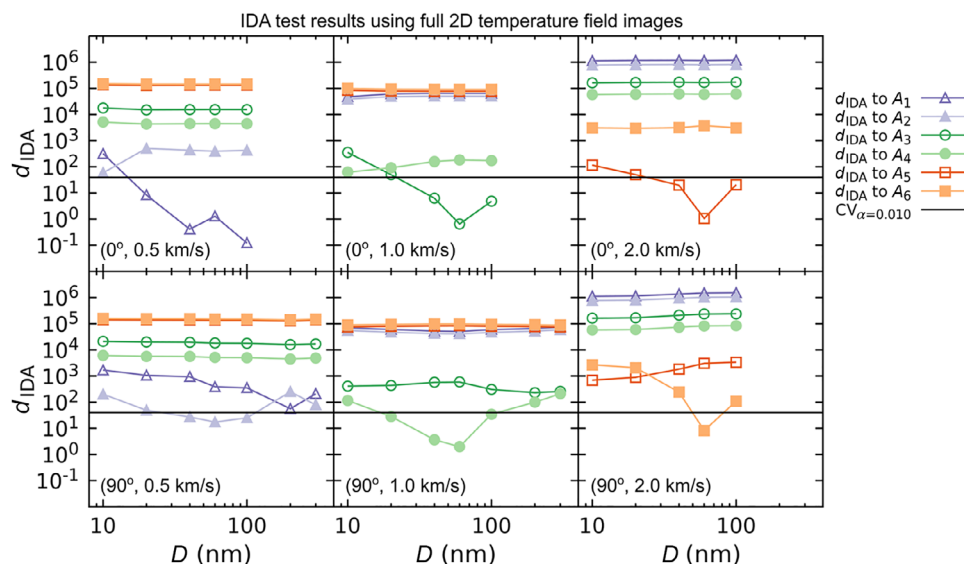


FIGURE 10 | Comparison of d_{IDA} distances to the six baseline shock conditions computed using full 2D images of temperature fields plotted as a function of pore diameter D at selected fixed θ and u_p . The d_{IDA} CV corresponding to 99.0% ($\alpha = 0.010$) confidence is plotted as a solid line.

Figure 10 shows the results of the IDA test in which each hot spot image is compared to the baselines established for the six different baseline conditions listed in Table 2. As an example, the top left-most panel shows results for comparing five different simulations performed for the $(0^\circ, 0.5 \text{ km/s})$ case with different diameters D . Each set of symbols and curves corresponds to the IDA distance d_{IDA} of these five simulations to one of the different shock conditions. Simulations for which d_{IDA} is smaller than the CV cannot be distinguished from the baseline.

Focusing first on the $\theta = 0^\circ$ results in the top row shows that hot spots produced for this shock orientation clearly exhibit scale-invariant features. This can be seen by comparing the open-symbol distances for conditions A_1 , A_3 , and A_5 against the CV at significance level $\alpha = 0.01$. For the weakest shock, all but the 10 nm pores exhibit intracondition d_{IDA} distances smaller than the CV. The IDA test reveals a potentially unintuitive result in that the diameter interval for scale invariance narrows as the shock strength increases. Both the 10 and 20 nm pore cases can be distinguished from their respective baselines when $u_p \geq 1.0 \text{ km/s}$.

IDA test results for the 90° simulations in the bottom row show similar results to those obtained for the 0° simulations. Both the $(90^\circ, 0.5 \text{ km/s})$ and $(90^\circ, 1.0 \text{ km/s})$ simulations with $D \in [20 \text{ nm}, 100 \text{ nm}]$ are indistinguishably close to their respective baselines (A_2 and A_4). While the IDA test indicates that the $D = 200 \text{ nm}$ and 300 nm simulations are distinguishable from the 60 nm baselines at $\alpha = 0.010$, these simulations are still quite similar and are indistinguishable from the baselines at somewhat lower α levels (0.001). One distinct difference between the 0° and 90° results is that the $(90^\circ, 2.0 \text{ km/s})$ simulations are generally further from their baseline (A_6) than the $(0^\circ, 2.0 \text{ km/s})$ simulations are from their baseline (A_5). In fact, the IDA test indicates that all three shock strengths yield hot spot structures for the 90° orientation that are further from their baselines than was seen for the 0° orientation.

The fact that the IDA test finds more sensitivity to D for the 90° orientation compared to the 0° orientation is likely because the 90° orientation promotes the formation of shear bands [43]. Two types of shear bands can be seen in the 90° simulation images, including those that emanate from the central core of the hot spot and those that form homogeneously in the bulk crystal when the shock strength is above the apparent Hugoniot elastic limit ($u_p > 0.5 \text{ km/s}$). This contrasts strongly with the crystal deformation modes that activate in the 0° simulations, which can include a largely nondissipative buckling/twinning mode [44] but no shear banding [43]. While the orientations of the shear bands coincide with known crystallographic slip systems that break the molecular layers, the precise nucleation points for these bands are at least partially stochastic [45].

The above tests clearly show that the hot spot temperature fields depend more strongly on u_p than θ . In all cases, simulations with a given u_p are closer to the two θ baselines established for that u_p than they are from any of the other baselines. For instance, the top left-most panel shows that the $(0^\circ, 0.5 \text{ km/s})$ simulations are generally closest to the $A_1 = (60 \text{ nm}, 0^\circ, 0.5 \text{ km/s})$ baseline and next closest to the $A_2 = (60 \text{ nm}, 90^\circ, 0.5 \text{ km/s})$ baseline. It is perhaps surprising that some of the smallest simulations for a given (θ, u_p) are closer to the alternative θ baseline than they are to their “correct” baseline, such as can be seen in the top left-most panel where the $(10 \text{ nm}, 0^\circ, 0.5 \text{ km/s})$ simulation is closer to A_2 than to A_1 .

Common practice for the quantitative analysis of hot spot temperature fields has largely centered on comparisons of temperature distributions and/or their CDFs and CCDFs. While direct comparisons of 2D images with the IDA test are potentially more capable of detecting structural differences, the IDA test is general and can also work with 1D distribution functions directly. The only requirement for applying the IDA test is that a consistent registration and rasterization scheme be used when vectorizing the images, which in the case of distribution functions can be met by the use of a consistent binning scheme. Figure 11 shows IDA

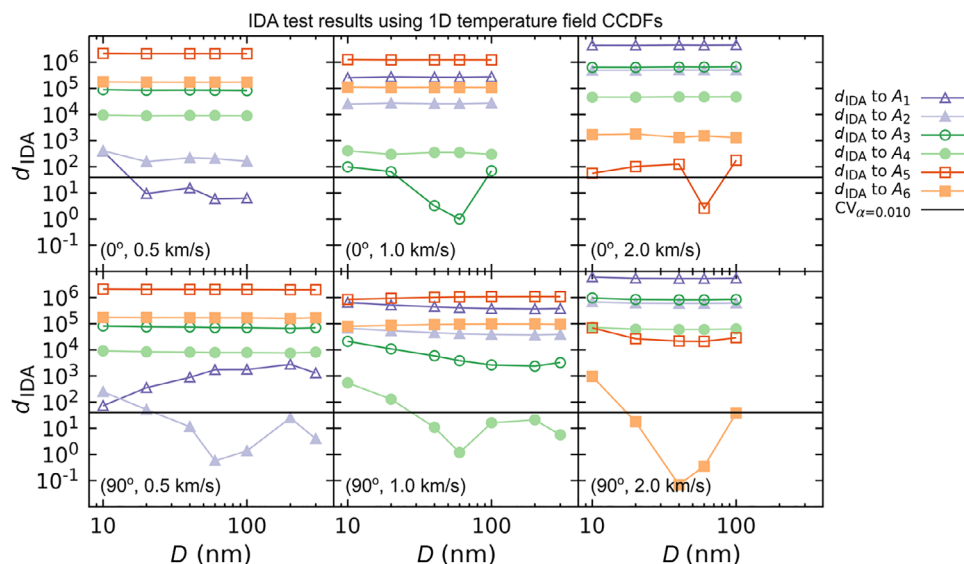


FIGURE 11 | Comparison of d_{IDA} distances to the six baseline shock conditions computed using the temperature field CCDFs plotted as a function of pore diameter D at selected fixed θ and u_p . The d_{IDA} CV corresponding to 99.0% ($\alpha = 0.010$) confidence is plotted as a solid line.

test results for the hot spot CCDFs computed using a bin width of $\Delta T = 1$ K. We note that essentially identical classification results were obtained with coarser bin widths up to $\Delta T = 50$ K.

Application of the IDA test to the temperature CCDFs yields results that are generally consistent with those obtained using the full 2D images, but with several key differences. First, the CCDFs for the 200 and 300 nm pores are indistinguishable from the 60 nm baseline. This provides a much stronger statistical demonstration that the 90° TATB hot spots exhibit scale-invariant distributions over at least the interval $D \in [40 \text{ nm}, 300 \text{ nm}]$ compared to our earlier inspection of these data [30]. Second, the CCDF-based IDA classification of the (0°, 2.0 km/s) simulations shows all other pore diameters are distinguishable from the 60 nm baseline whereas the image-based IDA classification shows that pores with $D \in [40 \text{ nm}, 100 \text{ nm}]$ are indistinguishable from the 60 nm baseline. This stems from the fact that the (60 nm, 0°, 2.0 km/s) baseline CCDFs cluster very tightly, so even small deviations become noticeably large. In this respect, the image-based IDA classification is more powerful as it accounts for similarities in the hot spot shape, not just the distribution of temperature values. This indicates that important discernible structural information present in the 2D images is lost when those images are reduced to 1D distribution functions. While this result is perhaps not surprising, it highlights the limitations of distinguishability tests that only operate on 1D distributions such as the KS test.

Several distinctions between the KS test and the IDA test warrant further discussion, especially as pertains to the application of the IDA test to 1D distribution data. First is that the KS test can be framed in a way that compares a test distribution to a known distribution. (This is distinct from the two-sided KS test applied in connection with Figure 1.) This could be achieved with the MD image data at hand by approximating the known distribution as a mean CDF computed over the baseline simulations. However, in this respect, the IDA test is superior in that it accounts for both the mean and variance of the baseline distributions. Second, and

arguably more important, is that the IDA test can be uniformly applied to a much broader class of descriptors than the KS test. In the present case, we focused on images as 2D scalar field data, but the IDA test can support data of arbitrarily large finite dimensionality.

In summary, applying the IDA test to the hot spot data at hand shows that the 1D temperature distributions are indistinguishable (i.e., scale invariant) relative to the 60 nm baseline over size ranges that vary depending on the shock condition (see Figure 11). The 0° cases considered pores up to 100 nm and the range of sizes exhibiting scale invariance narrows as the shock strength increases, from $D \in [20 \text{ nm}, 100 \text{ nm}]$ at 0.5 km/s, to $D \in [40 \text{ nm}, 60 \text{ nm}]$ at 1.0 km/s, and with all sizes being distinguishable from the 60 nm baseline at 2.0 km/s. The 90° cases considered pores up to 300 nm for the two weakest shock strengths and up to 100 nm for the strongest shock strength. Here, the two weakest cases exhibit scale invariance over $D \in [40 \text{ nm}, 300 \text{ nm}]$ and the strongest exhibits scale invariance over $D \in [20 \text{ nm}, 100 \text{ nm}]$. Application of the IDA test to the full 2D temperature fields yields somewhat different results (see Figure 10). For instance, in the (0°, 0.5 km/s) case, both the 1D distributions and 2D fields exhibit scale invariance over the same set of sizes $D \in [20 \text{ nm}, 100 \text{ nm}]$, but the two 0° cases with the strongest shocks exhibit invariance in the 2D fields over a larger size interval $D \in [40 \text{ nm}, 100 \text{ nm}]$ compared to the 1D distributions. In contrast, in the 90° cases, the 2D fields exhibit invariance over narrower size intervals compared to the 1D distributions and that invariance does not extend beyond 100 nm for any shock strength.

Three points warrant additional emphasis. First, it should be noted that because the IDA test is inherently comparative, it can only identify when trends (such as invariance) are present in the data at hand. It does not give a confidence factor on extrapolation outside the measured range. Second, our study highlights a potential pitfall in the quantitative analysis of hot spots. Qualitatively different conclusions regarding size ranges

over which hot spot scale invariance holds for a given shock condition can be reached depending on whether the hot spot is represented as a spatially correlated temperature field as opposed to a distribution function that lacks spatial information. Third, while the IDA test can determine whether a hot spot image is indistinguishable within the established variance of a baseline, it cannot determine whether that variance is sufficient to lead to qualitatively different pathways for hot spot evolution (e.g., reactive growth or thermal dissipation) unless the underlying image data capture those events. Our nonreactive MD simulations capture only the initial hot spot temperature field and cannot be used to predict chemically reactive evolution. Future work using a reactive model could explore whether the variance in the high-temperature distribution tails of the initial hot spots identified here (see Figure 9) is sufficient to induce qualitatively different evolution responses. The IDA test can be applied to image time series of hot spot evolution, not just the static images considered here, which would enable similarity comparisons of ensuing dynamical processes, not just the initial condition.

6 | Conclusions

Hot spot fields have historically been analyzed and compared in either a qualitative manner or through the lens of the temperature distribution, which neglects spatial correlations. Quantitative comparison of images resulting from evolution through different times or due to different length scales and boundary conditions can be a nuanced exercise in face of underlying stochastic variations and spatial correlations accompanying most systems. Standard algorithms for comparing image-like data, such as the KS test, work by reducing high-dimensional data to a 1D distribution function that eliminates potentially important spatial correlations. Such tests also have limitations in accounting for variance in images that for physical reasons are considered equivalent. On the other hand, PC-based techniques can account for spatial correlations, but often work in a very large feature space and employ ad hoc thresholding for image classification. To overcome these limitations, we develop a statistical approach that we call the IDA test, which works in a reduced PC space and incorporates a test statistic to calculate p -values to detect when images are distinguishable from an established baseline beyond stochastic variations.

The IDA test makes comparatively few assumptions regarding image data, but it does have two key requirements. First is that a few (e.g., 3+) sample images are required to quantify the expected mean and variance of a given baseline condition. Second is that a consistent pixel-by-pixel rasterization and registration scheme must be used for all the images. What constitutes an image is highly general and there is no requirement that the data be 2D in nature. Acceptable images can have lower or higher dimensionality (e.g., 0D scalar, 1D spectrum, or ≥ 3 D voxel grid), include multiple appended maps of scalar/vector fields, and even depict a connected series of snapshots such as a time history. While the second requirement may appear limiting, it can be easily overcome. The IDA test cannot be directly applied to compare images of a stochastic nature that lack consistent registration, but it can be applied to consistently sampled summary features. As an example, computed tomography images of a heterogeneous material microstructure containing

grains and pores are inherently stochastic [32], but the IDA test can be applied to summary features such as grain count or grain-boundary surface area.

The IDA test requires specifying two parameters in the distance metric d_{IDA} . One of these is the number of PCs r considered in the sum (Equation 12). Here, we recommend simply choosing $r = n - 1$, which is the full rank of the tensor $\mathbf{M}_c$ (see Equation 5). At the same time, we note that the presence of the PC weight factors f_j make the construction of d_{IDA} and its supported conclusions insensitive to any reasonable choice of r that captures the first few “important” PCs. Second, setting the weighting factors equal to the PC weights f_j was a choice made out of simplicity and may not be a unique selection backed by fundamental theoretical justification. We have not explored other more general choices such as f_j^ν , with exponent $\nu > 0$. However, as we have demonstrated in this work, our specific choice of weights (f_j with $\nu = 1$) leads to consistent classification of shock condition for all hot spot structures over a reasonable range of significance level α . This consistency is not achieved with the uniform weight choice $f_j = 1$, likely because the stochastic noise represented within high-index PCs does not get damped out.

One of the key steps in the IDA test is representing images in a reduced-dimensional space, which we perform using PCA. It is worth noting that there are other methods for reduced-dimensional representation of images, such as t-distributed stochastic neighbor embedding (t-SNE) [46]. This is a powerful technique that has multifaceted applications. However, unlike with PCA, reduced-dimensional representation with t-SNE is strongly dependent on parameter choices. Given the stochastic nature of such embedding, it is not suitable for measuring precise distances with well-defined criticality. The PC-based IDA distance metric proposed in this work is also just one of many possible choices of distance with well-defined statistical criticality. For instance, the stochasticity obtained from multiple runs within the same condition could also be expressed via stochasticity in distance of the test image from the stochastic replicas of the stated condition using comparison of either pixel-level intensities or intensity histograms. A variety of standard distance/similarity measures could be employed in this regard, including the cosine similarity, Euclidean, and various generalizations thereof for pixel-level data, and the KS metric or the Kullback–Liebler divergence for histogram-based distances [31, 47]. The challenge with each such measure is to uncover the underlying probability distribution so that the statistical significance (i.e., p -value) of the resulting distance can be properly quantified.

We anticipate that the IDA test can help in a wide variety of contexts in energetic materials research beyond characterizing hot spots, including characterizing off-normal changes in materials (e.g., aging-induced), model parameter sensitivity analyses and uncertainty quantification, coarse-grain and machine learning model validation, and in the direct comparison of computational models and experiments. In contexts where some images are easier to obtain than others (e.g., model validation), the IDA test can be used to determine whether a single hard-to-obtain sample is consistent with the variation evident in multiple easier-to-obtain samples. By framing comparisons in terms of statistical distinguishability rather than subjective inspections, the IDA test also provides an unambiguous definition of what constitutes good

agreement. A computational model might reasonably be considered *good enough* if the IDA test shows its high-dimensional outputs are indistinguishable from some baseline truth such as an experimental measurement or high-fidelity simulation.

Author Contributions

M.P.K. designed and performed the MD simulations and framed the analysis approach to test image distinguishability. A.M. recast the problem in the principal component space and formulated the distance metric for the IDA test. M.P.K. and A.M. iteratively refined the IDA test and wrote the manuscript. L.E.F. helped design the MD simulations and advised on high explosive shock physics and data analysis methods. All authors reviewed and approved the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: prep70140-sup-0001-SuppMat.pdf **Supporting File 2:** prep70140-sup-0002-SuppMat.zip